Summary on Simultaneous bandwidths determination for DK-HAC estimators and long-run variance estimation in nonparametric settings

Thapakorn PIPATPAJONG, Liangbang LI

July 2023

1 Background, Settings, and Challenges

1.1 Background

Long-run variance (LRV) plays a key role for heteroscedasticity and autocorrelation robust (HAR) inference. The classical approach in HAR inference relies on consistent estimation of the LRV.

Newey and West (1987) and Andrews (1991) proposed kernel heteroscedasticity and autocorrelation consistent (HAC) estimators:

$$\hat{J}_{Cla,T} = \hat{J}_{Cla,T} \triangleq \frac{T}{T-p} \sum_{k=-T+1}^{T-1} K_1(b_{1,T}k)\hat{\Gamma}_{Cla}(k), \quad \text{with}$$

$$\hat{\Gamma}_{Cla}(k) \triangleq \begin{cases} T^{-1}\sum_{t=k+1}^{T} \hat{V}_t\hat{V}'_{t-k}, & k \geq 0 \\ T^{-1}\sum_{t=-k+1}^{T} \hat{V}_{t+k}\hat{V}'_t, & k < 0 \end{cases}$$

To flexibly account for nonstationarity, Casini (2021) introduced a double kernel HAC (DK-HAC) estimator that applies kernel smoothing over two directions. The first one smooths the lagged autocovariances, similar to the classical work. The second one applies smoothing over time, which is better for nonstationary models. Also, Casini (2021) proposed the bandwidths selection by selecting bandwidths sequentially where changing one bandwidth will affect another one, but not vice versa.

In this paper, the authors consider simultaneous bandwidths determination obtained by jointly minimizing the global MSE of the DK-HAC estimator. They obtain the asymptotic optimal formula for the two bandwidths and use the plug-in approach to replace unknown quantities by consistent estimates.

1.2 Settings

LRV is defined by $J \triangleq \lim_{T\to\infty} J_T$ where $J_T = T^{-1}\sum_{s=1}^{T}\sum_{t=1}^{T}\mathbb{E}(V_s(\beta_0)V_t(\beta_0)')$. The LRV can be estimated by using DK-HAC estimator:

$$\hat{J}_T = \hat{J}_T(b_{1,T}, b_{2,T}) \triangleq \frac{T}{T-p} \sum_{k=-T+1}^{T-1} K_1(b_{1,T}k)\hat{\Gamma}(k), \quad \text{with}$$

$$\hat{\Gamma}(k) \triangleq \frac{n_T}{T-n_T} \sum_{r=0}^{\lfloor (T-n_T)/n_T \rfloor} \hat{c_T}(rn_T/T, k), \quad \text{with}$$

$$\hat{c_T}(rn_T/T, k) \triangleq \begin{cases} (Tb_{2,T})^{-1}\sum_{s=k+1}^{T} K_2^*\left(\frac{((r+1)n_T-(s-k/2))/T}{b_{2,T}}\right)\hat{V}_s\hat{V}'_{s-k}, & k \geq 0 \\ (Tb_{2,T})^{-1}\sum_{s=-k+1}^{T} K_2^*\left(\frac{((r+1)n_T-(s+k/2))/T}{b_{2,T}}\right)\hat{V_{s+k}}\hat{V}'_s, & k < 0 \end{cases}$$

Where the kernels K_1 and K_2 have the following properties:

$$K_1 = \Big\{ K_1(\cdot):\ \mathbb{R}\to[-1,1]:\ K_1(0)=1,\ K_1(x)=K_1(-x),\ \forall x\in\mathbb{R}$$
$$\int_{-\infty}^{\infty} K_1^2(x)dx < \infty,\ K_1(\cdot) \text{ is continuous at 0 and at all but finite numbers of points} \Big\}.$$

$$K_2 = \Big\{ K_2(\cdot) : \mathbb{R} \to [0,\infty] :\ K_2(x) = K_2(1-x),\ \int K_2(x)dx = 1$$
$$K_2(x) = 0,\ \text{for } x \notin [0,1],\ K_2(\cdot) \text{ is continuous} \Big\}.$$

1.3 Challenges

From the DK-HAC estimation, we need to find the optimal bandwidth $(b_{1,T}, b_{2,T})$ for accurate estimation. In this article, the criteria to determine the optimal bandwidths are called global asymptotic relative mean squared error (ReMSE), which defined as:

$$\lim_{T\to\infty} \text{ReMSE}\left(Tb_{1,T}b_{2,T},\ \hat{J}_T(b_{1,T}, b_{2,T}J^{-1}),\ W_T\right)$$
$$= \lim_{T\to\infty} Tb_{1,T}b_{2,T}\mathbb{E}\left(\text{vec}\left(\hat{J}_T J^{-1} - I_p\right)' W_T \text{vec}\left(\hat{J}_T J^{-1} - I_p\right)\right)$$

In this article, the selected bandwidths are called simultaneous bandwidth. The bandwidths are dependent with each other. Changing $b_{1,T}$ will affect $b_{2,T}$ and vice versa. The goal is to minimize the ReMSE by choosing optimal bandwidths and kernels for the estimation of LRV in both semiparametric and nonparametric settings.

2 Ideas, Implementations, Theories, Pros and cons

2.1 Ideas

In previous work of Casini's (2021), the bandwidth $b_{2,T}$ is determined first, then the bandwidth $b_{1,T}$ will be computed after. Changing $b_{2,T}$ will affect $b_{1,T}$ but not vice versa. Therefore, the downside of sequential bandwidths is that it will likely not minimize the global MSE because $b_{1,T}$ and $b_{2,T}$ do not affect with each other. Therefore, the simulataneous bandwidths are introduced in this paper under the assumption that $\{V_{t,T}\}$ is segmented locally stationary (SLS) with mean-zero process.

2.2 Implementations

This paper proposed the implementations of automatic data-dependent bandwidths. The first step is to specify p univariate parametric models for $V_t = (V_t^{(1)}, V_t^{(2)}, \ldots, V_t^{(p)})$. The second step is to estimate the local least squares (LS) by using local data to the left of $u = t/T$:

$$\hat{a}_1^{(r)}(u) = \frac{\sum_{j=t-n_{2,T}+1}^{t} \hat{V}_j^{(r)}\hat{V}_{j-1}^{(r)}}{\sum_{j=t-n_{2,T}+1}^{t}\left(\hat{V}_{j-1}^{(r)}\right)^2}, \quad \hat{\sigma}^{(r)}(u) = \left(\sum_{j=t-n_{2,T}+1}^{t}\left(\hat{V}_j^{(r)} - \hat{a}_1^{(r)}(u)\hat{V}_{j-1}^{(r)}\right)^2\right)^{1/2}.$$

The next step is to estimate $\bar{\Delta}_{1,2,1} \triangleq \sum_{k=-\infty}^{\infty}\int_0^1 (\partial^2/\partial u^2)c(u,k)\,du$ by estimate $\Delta_{1,2,1}^{(r,r)}(u,k)$ denoted as $\hat{\Delta}_{1,2,1}^{(r,r)}(u,k)$, then estimate $\bar{\Delta}_{1,2,1}$ denoted as $\hat{\bar{\Delta}}_{1,2,1}^{(r,r)}$ afterwards.

The last step is to estimate the bandwidths $\hat{b}_{1,T} = 0.46\hat{\phi}_1^{1/24}T^{-1/6}$ and $\hat{b}_{2,T} = 3.56\hat{\phi}_2^{1/24}T^{-1/6}$ where $\hat{\phi}_1 = \hat{\phi}_{1,1}/\hat{\phi}_{1,2}^5$, $\hat{\phi}_2 = \hat{\phi}_{1,2}/\hat{\phi}_{1,1}^5$,

$$\hat{\phi}_{1,1} = (4\pi)^{-2}\sum_{r=1}^{p} W^{(r,r)}\left(\hat{\bar{\Delta}}_{1,2,1}^{(r,r)}\right)^2 / \left(\frac{n_{3,T}}{T}\sum_{j=0}^{\lfloor T/n_{3,T}\rfloor - 1}(\hat{\sigma}^{(r)}(jn_{3,T}+1))^2(1-\hat{a}_1^{(r)}(jn_{3,T}+1))^{-2}\right)^2,$$

$$\hat{\phi}_{1,2} = 36\sum_{r=1}^{p} W^{(r,r)}\left(\frac{\sum_{j=0}^{\lfloor T/n_{3,T}\rfloor-1}((\hat{\sigma}^{(r)}(jn_{3,T}+1))^2\hat{a}_1^{(r)}(jn_{3,T}+1))(1-\hat{a}_1^{(r)}(jn_{3,T}+1))^{-4}}{\sum_{j=0}^{\lfloor T/n_{3,T}\rfloor-1}(\hat{\sigma}^{(r)}(jn_{3,T}+1))^2(1-\hat{a}_1^{(r)}(jn_{3,T}+1))^{-2}}\right)^2.$$

We can set $n_{3,T} = n_{2,T} = n_T$ and $W^{(r,r)}$ is an $p \times p$ weighted diagonal matrix, the usual choice is $W^{(r,r)} = 1$ for all r.

2.3 Theories

The theories are based on the assumptions about the stochastic process $\{V_{t,T}\}_{t=1}^{T}$ that it is Segmented Locally Stationary (SLS) process with mean 0. From the asymtotic MSE properties, both $b_{1,T}$ and $b_{2,T}$ will affect bias and variance which will be useful in jointly minimize ReMSE

The rate of converge of the estimator $\hat{J}_T$ can be represented as $\sqrt{Tb_{1,T}b_{2,T}}(\hat{J}_T - J_T) = O_{\mathbb{P}}(1)$. Therefore, the rate of convergence of $\hat{J}_T$ is $\sqrt{Tb_{1,T}b_{2,T}}$, which is the same as the one in the sequential bandwidths.

The optimal kernels K_1 and K_2 are given by

$$K_1^{opt} = K_1^{QS} = \frac{25}{12\pi^2 x^2}\left(\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5)\right),$$

$$K_2^{opt}(x) = 6x(1-x), \text{ for } x \in [0,1].$$

Both optimal bandwidths $(\tilde{b}_{1,T}, \tilde{b}_{2,T})$ discussed in implementations section have the order of $O(T^{-1/6})$ while the sequential bandwidths have $b_{1,T}^{opt} = O(T^{-4/25})$ and $\bar{b}_{2,T}^{opt} = O(T^{-1/5})$. The sequential bandwidths are better for nonstationary process because the growth rate of $\bar{b}_{2,T}^{opt}$ is slower than $\tilde{b}_{2,T}$ and smaller block length will be beneficial for smoothing over time. However, the rate of convergence of simultaneous bandwidths is $O(T^{1/3})$ which is slightly faster than the sequential bandwidths $O(T^{0.32})$.

The relationship between $\tilde{b}_{1,T}$ and $\tilde{b}_{2,T}$ is that when there are more variation around the zero frequency, $\tilde{b}_{1,T}$ will decrease for better smoothing for frequency while $\tilde{b}_{2,T}$ decreases. On the other hand, If there are more nonstationary data, $\tilde{b}_{2,T}$ will decrease for better smoothing over time while $\tilde{b}_{1,T}$ increases.

2.4 Pros and cons

2.4.1 Pros

1. The author uses the plug-in approach to construct data-dependent bandwidths, which is more reasonable, compared with the sequential criterion.
2. joint MSE criterion performs better in terms of size control, especially when the data is stable.
3. The HAR tests normalized by DK-HAC estimators strike the best balance between empirical size and power among the existing LRV estimators.

2.4.2 Cons

1. When the nonstationarity is high, the sequential criterion performs marginally better.
2. This is like a 'seesaw', it's impossible to be the best at the same time. The joint criteria has better in variance and the sequential criteria has better control in bias.
3. The estimator is sensitive to outliers or extreme values, which may give inaccurate results.

3 Improvement

3.1 Our Newly Proposed Estimator

We focus on improve the estimator by make it more robust to the outliers. Instead of using an average, we use median to neglect the outliers while keeping the concepts of the estimator the same. The main idea is to change every average equation into median instead and make some adjustments. Therefore, the newly proposed DK-HAC estimator is

$$\hat{J}_T = \hat{J}_T(b_{1,T}, b_{2,T}) \triangleq \frac{T}{T-p}\sum_{k=-T+1}^{T-1} K_1(b_{1,T}k)\hat{\Gamma}(k), \quad \text{with}$$

$$\hat{\Gamma}(k) \triangleq T^{1/6}\left(\frac{n_T}{T-n_T}\right)\sum_{r=0}^{\lfloor (T-n_T)/n_T \rfloor} \hat{c_T}(rn_T/T, k), \quad \text{with}$$

$$\hat{c_T}(rn_T/T,k) \triangleq \begin{cases} \text{median}(\hat{V}_s\hat{V}'_{s-k},\ k+1\le s\le T)(Tb_{2,T})^{-1}\sum_{s=k+1}^{T} K_2^*\left(\frac{((r+1)n_T-(s-k/2))/T}{b_{2,T}}\right), & k\ge 0 \\ \text{median}(\hat{V_{s+k}}\hat{V}'_s,\ k+1\le s\le T)(Tb_{2,T})^{-1}\sum_{s=-k+1}^{T} K_2^*\left(\frac{((r+1)n_T-(s+k/2))/T}{b_{2,T}}\right), & k<0 \end{cases}$$

The difference on this estimator is that we use only the median of $\hat{V}_s\hat{V}'_{s-k}$ or $\hat{V_{s+k}}\hat{V}'_s$. However, this will lead to great inaccuracy. Therefore, we multiply the equation by $T^{1/6}$ on $\hat{\Gamma}(k)$ by our simulation, which gives the same performance as the proposed estimator in the article.

The data-dependent bandwidths have the similar equations, except $\hat{\phi}_{1,1}$ and $\hat{\phi}_{1,2}$, which change to

$$\hat{\phi}_{1,1} = (4\pi)^{-2}\sum_{r=1}^{p} W^{(r,r)}\left(\hat{\bar{\Delta}}^{(r,r)}_{1,2,1}\right)^2 \Big/ \left(\text{median}\left[(\hat{\sigma}^{(r)}(jn_{3,T}+1))^2(1-\hat{a}_1^{(r)}(jn_{3,T}+1))^{-2},\ 0\le j\le \lfloor T/n_{3,T}\rfloor -1\right]\right)^2,$$

$$\hat{\phi}_{1,2} = 36\sum_{r=1}^{p} W^{(r,r)}\left(\frac{\text{median}\left[((\hat{\sigma}^{(r)}(jn_{3,T}+1))^2\hat{a}_1^{(r)}(jn_{3,T}+1))(1-\hat{a}_1^{(r)}(jn_{3,T}+1))^{-4},\ 0\le j\le \lfloor T/n_{3,T}\rfloor -1\right]}{\text{median}\left[(\hat{\sigma}^{(r)}(jn_{3,T}+1))^2(1-\hat{a}_1^{(r)}(jn_{3,T}+1))^{-2},\ 0\le j\le \lfloor T/n_{3,T}\rfloor -1\right]}\right)^2.$$

3.2 Simulation Results

We simulate the results by estimate AR(1) models with different sample sizes $T = 100, 200, 400,$ and 800. We compare the log mean squared error between the article estimator and the newly proposed one. The results shows that our proposed estimator is slightly more accurate and consistent than the estimator proposed in article under the normal settings, as shown in figure 1. Under the big jumps settings, our proposed estimator performs the same as normal settings while the one proposed in article has higher errors, as shown in figure 2.

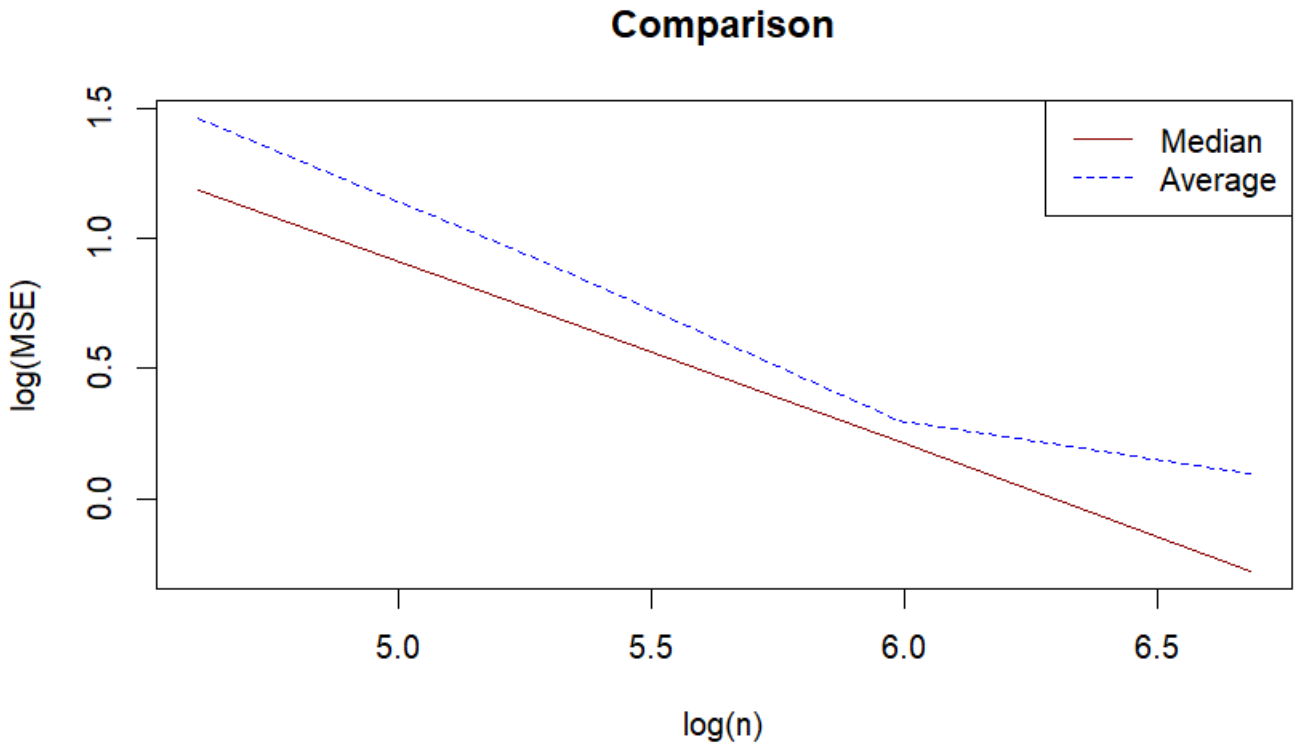


Figure 1: The comparison between our proposed estimator and the one proposed in article under normal settings.

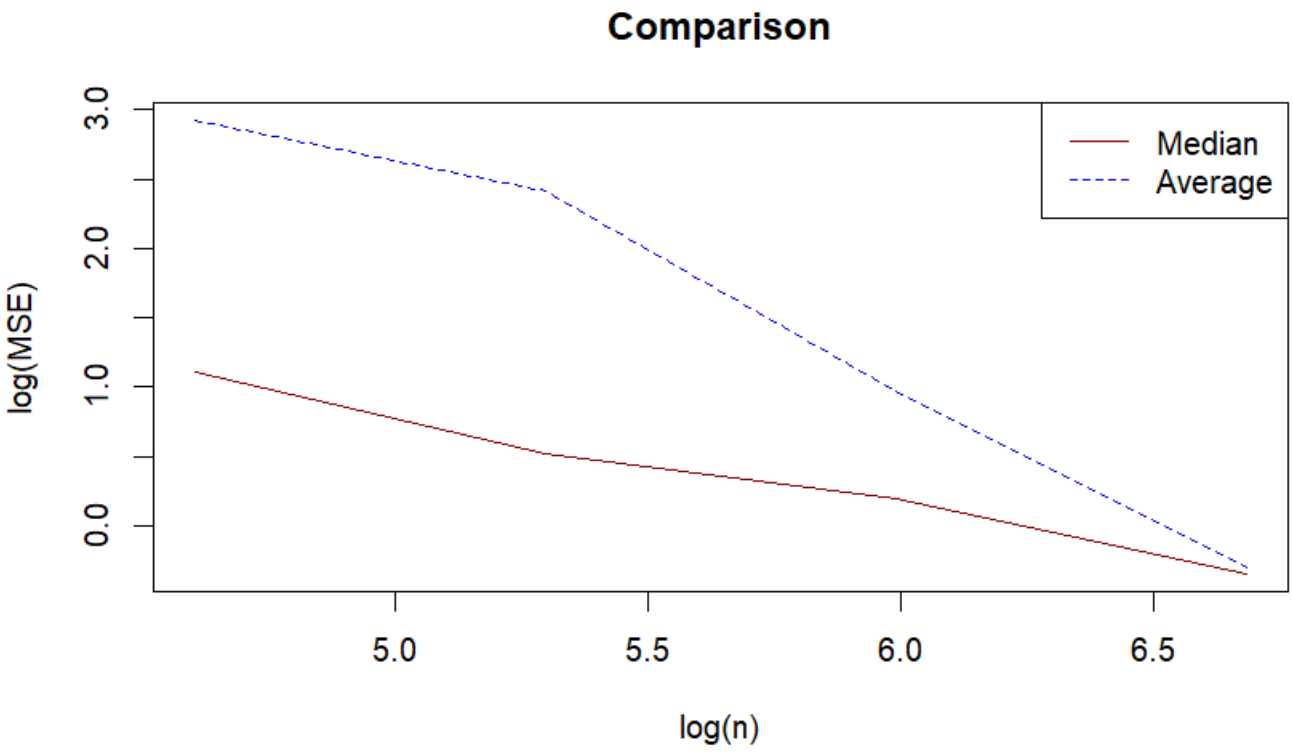


Figure 2: The comparison between our proposed estimator and the one proposed in article under time series with jumps settings.

4 What have we done

1. Summarize the article.
2. Translate the formulas in the article into R codes.
3. Write the R codes of the sequential criteria for optimal band-widths and compare these two.
4. Write the R codes of the prewhiting steps.
5. Use the codes try to reproduce the result in section 6.
6. Try giving a proposal to improve the method.